

VERIDIAN BANK

SentinelAI Fraud Detection Platform

PROJECT CHARTER

SentinelAI Fraud Detection Platform

Real-Time, ML-Driven Card Fraud Detection & Decisioning

Document Owner:	Mautik Patel, Project Delivery Manager
Sponsor:	Sarah Whitfield, Chief Risk Officer
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Classification:	Confidential – Internal Use Only

1. Project Purpose & Justification

Veridian Bank's card fraud losses have grown 42% over the past 24 months, driven primarily by card-not-present (CNP) e-commerce fraud and account takeover schemes. The current fraud detection engine relies on static, rules-based logic that was last materially updated in 2021. This has produced two compounding problems: fraud losses of approximately \$18.4M in the trailing twelve months, and a false-positive rate of 6.8% that generates over 40,000 unnecessary transaction declines per month, directly harming customer experience and contributing to a measurable increase in card attrition.

SentinelAI will replace the legacy rules engine with a real-time, machine-learning-based fraud scoring platform. The project is sponsored by the Office of the Chief Risk Officer and is one of three initiatives designated as a 2026 enterprise priority under the Bank's Risk Modernization Program.

2. Business Objectives & Success Criteria

Objective	Success Criteria	Target Date
Reduce fraud losses	Reduce annualized fraud losses by $\geq 35\%$ vs. FY2025 baseline	Q4 2026
Reduce false positives	Cut false-positive decline rate from 6.8% to $\leq 4.0\%$	Q4 2026
Real-time performance	Achieve $\leq 200\text{ms}$ P95 transaction scoring latency	Go-live
Analyst productivity	Reduce manual case review volume by 30%	Q1 2027 (90 days post go-live)
Regulatory alignment	Pass independent model risk validation (SR 11-7 aligned)	Prior to go-live

3. Scope

3.1 In Scope

- Real-time ML fraud scoring engine for credit and debit card transactions (POS and CNP)
- Feature engineering pipeline using Snowflake as the analytical data layer
- Model development, validation, and champion/challenger testing framework
- Case management workflow integration for the Fraud Operations team
- Integration with the existing card authorization switch (real-time API)
- Explainability layer (reason codes) for analyst review and regulatory response
- Model monitoring, drift detection, and retraining pipeline

3.2 Out of Scope

- Wire transfer and ACH fraud detection (separate initiative, Treasury Fraud Program)
- Merchant-side fraud/chargeback dispute automation
- Replacement of the core card issuance or authorization switch platform
- New customer-facing fraud alert mobile app (Phase 2 candidate)

4. Key Stakeholders

Name / Role	Organization	Interest / Influence
Sarah Whitfield, CRO	Risk Management	Executive Sponsor; accountable for fraud loss KPIs
Mautik Patel	PMO / BI Delivery	Project Delivery Manager; owns plan, budget, and delivery
David Okonjo, Head of Fraud Ops	Fraud Operations	Primary business owner of case workflow and analyst tooling
Priya Nataraj, Head of Data Science	Data & Analytics	Owns model design, training, and validation
Marcus Webb, Head of Card Payments	Card Payments	Owns authorization switch integration and uptime SLAs
Legal & Compliance	Compliance	Regulatory sign-off; model risk governance (SR 11-7)
Enterprise IT Infrastructure	IT	Provides compute, network, and production environment support

5. High-Level Milestones

Milestone	Target	Owner
Charter approval & funding release	Jan 2026	CRO / Sponsor
Requirements sign-off (BRD/FRD)	Feb 2026	Mautik Patel
Data pipeline & feature store live (dev)	Apr 2026	Data Engineering
Model v1 trained & validated	Jun 2026	Data Science
Champion/challenger pilot (10% traffic)	Jul 2026	Fraud Ops / Data Science
Model risk validation sign-off	Aug 2026	Compliance
Full production go-live	Sep 2026	Mautik Patel
Post-implementation review	Dec 2026	Mautik Patel

6. Budget Summary

Category	Amount (USD)	% of Total
Data science & model development (staff)	\$820,000	34%
Engineering & integration (staff)	\$610,000	25%
Cloud/Snowflake compute & storage	\$340,000	14%
Third-party fraud data & model tooling licenses	\$280,000	12%
Model risk validation (external)	\$150,000	6%

Category	Amount (USD)	% of Total
Contingency (9%)	\$200,000	9%
Total	\$2,400,000	100%

7. Assumptions & Constraints

7.1 Assumptions

- Historical transaction data (24+ months) is available and of sufficient quality in the existing data warehouse
- Fraud Operations will allocate two senior analysts to the pilot for labeling and feedback
- The card authorization switch vendor can support a new real-time scoring API call within existing latency budgets

7.2 Constraints

- Go-live must occur before the November – January peak fraud season
- Model must be explainable to satisfy Reg E dispute-response and fair-lending requirements
- No downtime permitted on the production card authorization path during cutover

8. Approval

Name / Role	Signature	Date
Sarah Whitfield – Chief Risk Officer (Sponsor)		Jan 12, 2026
Mautik Patel – Project Delivery Manager		Jan 12, 2026
David Okonjo – Head of Fraud Operations		Jan 12, 2026